MIL-C-46995C (MU)
4 April 1969
SUPERSEDING
MIL-C-46995B (MU)
11 December 1964

MILITARY SPECIFICATION

CARTRIDGE, 81MM, HE, M374
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one type of HE cartridge designated as M374 for use with the 81MM Mortar.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-P-46568 - Primer, Percussion, M71A1E1, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1315

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DRAWINGS

ARMY

8881026 - Cartridge, 81MM, HE, M374, w/Fuze, PD, M524
9230175 - Container, Ammunition, Fiber, M252A3 for Cartridge 81MM M374, HE and M375, WP, in Jungle Wrap
9230176 - Box, Packing, Ammunition for 81MM Mortar Ammunition in Fiber Container, M252A3

PUBLICATIONS

ARMY

ET-8881026 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material

3.1.1 General.-Material and parts shall be in accordance with applicable specifications and drawings.

3.1.2 Clean scrap.-Clean scrap (see 6.7) produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for configuration. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Loaded cartridge.-The loaded cartridge shall comply with all requirements specified on Drawing (dwg.) 8881026 and with all requirements specified in applicable specifications.

3.3 Cavitation in high explosive charge (see 4.3.1)

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3.3.1 General.-Cavities having a maximum (max.) projected dimension of 3/32 inch or less shall be disregarded provided that they are separated by 1/2 inch or more of solid explosive.

3.3.2 Max. projected areas and dimensions of cavities.-The aggregate of the projected area of all cavities shall not exceed 1 square inch and no individual cavity shall exceed 1/2 square inch. Cavities located in the area of the charge which extends from the base of the fuze well to 2.5 inches below the base of the fuze well shall not exceed 3/4 inch in any direction and their aggregate projected area shall not exceed 5/8 square inch.

3.3.3 Piping cavity.-A piping cavity on the longitudinal axis of the projectile shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/4 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.3.4 Surface cavitation.-Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The max. dimension of any visible cavity in the faced (bottom) surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/16 inch or less shall be disregarded in this computation.)

3.3.5 Cracked charges and annular rings.-Not more than four transverse cracks shall be permitted in any charge. The cracks shall be no wider than 1/32 inch. Longitudinal cracks shall be disregarded. The presence of annular rings with a projected width of 1/16 inch max. and extended from the shell wall 1/4 inch max. into the charge shall not be cause for rejection.

3.3.6 Base separation.-The base separation from the inside of the rear of the shell to the base of the explosive shall be not more than 1/32 inch.

3.4 Proving ground test.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

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3.4.1 Failure to fire.-The cartridge shall not fail to fire.

3.4.2 Safety.-There shall be no premature burst or indication of metal parts separation in the mortar tube or in flight when fired at any charge zone.

3.4.3 Impact functioning.-The cartridge shall function upon impact. The fuze shall be set for super quick function.

3.4.4 Horizontal range

3.4.4.1 No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.4.4.2 The standard deviation in horizontal range shall be no greater than the maximum lot values shown in Table I for the increment charge indicated.

TABLE I

Increment Charge Zone	Max. Lot Standard Deviation (Meters)
1	20
9	50

3.5 Workmanship

3.5.1 General.-All loading and assembling operations of the cartridge shall be correctly performed in a thorough workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, sharp edges, unblended radii, grease, rust and foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.5.2 Loading of high explosive.-The interior surface of the projectile shall be dry and free of foreign objects when the high explosive is poured into the projectile. Solid explosive shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.5.3 Threads.-The mating surfaces of the threads shall be free of caked explosives and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care should be taken to prevent the chemical cleaner from coming into contact with the charge and becoming trapped in the projectile.

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3.5.4 Exterior surface.-After loading, the exterior of the cartridge shall be clean and free of explosive. The painting and marking shall be restored by the loading and assembling contractor on any component furnished to him painted or marked when such painting or marking has been marred or obliterated.

3.6 Initial Production Test.-The cartridges shall function satisfactorily in accordance with the requirements of 3.4 after being subjected to the sequence of environmental tests in 4.2.3.7.

3.7 Strength of heat weld.-Obturating ring heat weld shall not break upon application of ten pounds minimum load but shall break upon application of twenty-five pounds maximum load when tested as specified in paragraph 4.3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specification referenced in any of the contractual documents.

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f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of cartridges shall contain:

a. Metal parts or metal parts assemblies of one lot interfix number from one manufacturer.
b. Loaded fuzes, of one lot interfix number, from one manufacturer.
c. Loaded ignition cartridges, of one lot interfix number, from one manufacturer.

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d. Fin assemblies, of one lot interfix number, from one manufacturer.
e. Loaded primers, of one lot interfix number, from one manufacturer.
f. Propellant increments, made from not more than one lot of propellant, from one manufacturer.
g. Composition B of one lot interfix number from one manufacturer.
h. Obturating rings, of one lot interfix number, from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects when applicable) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Body loading assembly, prior to loading (see dwg. 8881025 covering a detail of dwg. 8881026).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Foreign object in charge cavity	Visual	01001
Major:	AQL 0.40 percent		
101.	Interior protective coating with bare spot or blister ...	Visual	01002
Minor:	AQL 1.0 percent		
201.	Masking tape missing	Visual	01003
202.	Evidence of poor workmanship (see 3.5)	Visual	01004

4.2.2.2 Body loading assembly, prior to assembly of liner (see dwg. 8881025 covering a detail of dwg. 8881026)

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major:	AQL 0.65 percent		
101.	Depth of fuze well, min. . . .	Gage	02001
102.	Diameter of fuze well, min . .	Gage	02002
103.	Foreign matter in fuze well. .	Visual	02003

Minor:	AQL 2.5 percent		
201.	Depth of fuze well, max. . . .	Gage	02004
202.	Diameter of fuze well, max . .	Gage	02005
203.	Surface cavitation excessive (see 3.3.4).	Visual	02006
204.	Obturating band area covered with masking tape.	Visual	02007
205.	Evidence of poor workmanship (see 3.5).	Visual	02008

4.2.2.3 Cartridge assembly prior to assembly of fuze (see dwg. 8881026).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined.

Major:	AQL 0.40 percent		
101.	Depth to bottom of liner, min.	Gage	03001

Minor:	AQL 2.5 percent		
201.	Liner missing or damaged . . .	Visual	03002
202.	Explosive in threads	Visual	03003
203.	Foreign matter in liner cavity including explosive.	Visual	03004
204.	Liner removable by hand. . . .	Manual	03005
205.	Pettman cement or shellac varnish, as applicable, missing from threads	Visual	03006
206.	Evidence of poor workmanship (see 3.5).	Visual	03007

4.2.2.4 Plate, pressure (see dwg. 9218640 covering a detail of dwg. 8881026).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined.

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Major:	AQL 1.0 percent		
101.	Diameter of counterbore	Gage	09001
102.	Outside diameter, min.	Gage	09002
103.	Total thickness	Gage	09003
104.	Web thickness	Gage	09004
105.	Concentricity of counterbore with outside diameter	Gage	09005
Minor:	AQL 0.65 percent		
201.	Outside diameter, max.	Gage	09006

4.2.2.5 Cartridge (see dwg. 8881026)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer head above flush with rear face of fin relief	Straight-Edge	04001
2.	Ignition cartridge missing or inverted	Visual	04002
3.	Fuze safety wire missing or insecure	Visual-Manual	04003
4.	Fin damaged or distorted	Visual	04004
5.	Warning label missing or unidentifiable	Visual	04005
6.	Propellant increments cut, torn or missing	Visual	04025
7.	Obturating ring missing	Visual	04030
Major:	100 percent inspection		
101.	Obturating ring not welded	Visual	04031
102.	Obturating ring not flush or below bourrelet diameter	Visual	04032
103.	Primer flash hole blockage (see 4.2.4)	Test	04035
104.	Forward increment holder loose	Manual	04038
Major:	AQL 1.5 percent		
105.	Total length, max.	Gage	04006
106.	Obturating band damaged	Visual	04007
107.	Any component damaged or incorrectly assembled	Visual	04008
108.	Fin assembly fails to meet disassembly torque	Test	04009
109.	Fuze fails to meet disassembly torque	Test	04010
110.	Pressure plate missing or inverted (prior to assembling fin assembly to body loading assembly)	Visual	04019

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111.	Thread sealing compound does not cover a minimum of one half the number of male shell body threads for a full 360 degrees (prior to asembly)	Visual	04023
112.	Thread sealing compound does not cover a minimum of one half the number of a male primer threads for a full 360 degrees (prior to assembly)	Visual	04024
113.	Primer fails to withstand reapplication of minimum assembly torque	Test	04029
114.	Protective bag assembly tape not secured to shell and bag for full 360 degrees	Visual	04026
115.	Fuze not fully seated	Visual	04013
116.	Sealing compound bead does not cover crimp on red end of ignition cartridge or mating surface on bottom of ignition cartridge cavity in housing (as applicable) for a full 360 degrees (prior to insertion of ignition cartridge into fin cavity)	Visual	04036
117.	Sealing compound bead does not cover crimp on yellow end of ignition cartridge or inside diameter of pressure plate counterbore (as applicable) for a full 360 degrees (prior to insertion of ignition cartridge into fin cavity)	Visual	04037

Minor: AQL 1.5 percent

201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	04011
202.	Marking misleading, missing or unidentifiable, including incorrect color code	Visual	04012
203.	Evidence of poor workmanship (see 3.5)	Visual	04014

4.2.2.6 Unsealed fiber container (see dwg. 9230175)

Categories	Defects	Method of Inspection	Code No.

Critical: None defined.

Major: AQL 0.65 percent

101.	Packing stop missing	Visual	06001
102.	Filler disc assembly missing	Visual	06002
103.	Desiccant bag missing	Visual	06003

Minor: None defined.

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4.2.2.7 Sealed fiber container (see dwg. 9230175)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Tape incomplete, badly wrinkled or missing	Visual	07001
102.	Container damaged	Visual	07002
Minor:	AQL 1.0 percent		
201.	Marking missing, misleading or unidentifiable	Visual	07003
202.	Contents loose	Visual	07004

4.2.2.8 Fiber container wrapped in barrier material prior to waxing - Jungle Wrap (see dwg. 9230175)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Barrier material improperly assembled	Visual	08001
102.	Barrier material ends improperly folded or taped	Visual	08002
Minor:	AQL 1.5 percent		
201.	Marking missing or incomplete	Visual	08003
202.	Gum label missing	Visual	08004
203.	Tear tab missing	Visual	08005
204.	Tear tab length improper	Gage	08006

4.2.2.9 Fiber container after waxing - Jungle Wrap (see dwg. 9230175)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wax coating incomplete or damaged	Visual	05001

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Minor:	AQL 1.0 percent		
201.	Marking missing or unidentifiable	Visual	05002
202.	Kraft paper improperly placed on container	Visual	05003

4.2.2.10 Sealed wood packing box (see dwg. 9230176).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Contents exposed	Visual	10001
102.	DOD symbol missing or incorrect.	Visual	10002
103.	Board broken or split.	Visual	10003
104.	Hardware or strapping missing, broken or loose.	Visual-Manual	10004
Minor:	AQL 2.50 percent		
201.	Marking misleading, missing or unidentifiable	Visual	10005
202.	Hardware improperly engaged. . .	Visual-Manual	10006
203.	Car seal missing, unsealed or improperly positioned.	Visual-Manual	10007
204.	Contents loose	Visual-Manual	10008
205.	Handle missing or insecure . . .	Visual-Manual	10009

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of Primer, Percussion, M71A1E1 (see Specification MIL-P-46568), Code No. 03008.-If the total elapsed time between original acceptance of the M71A1E1 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M71A1E1 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46568 immediately before the M71A1E1 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.5).

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4.2.3.2 Cavitation (see 3.3 except 3.3.4) Major defect - Code No. 02009.-The sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-1235 using a CSP-2 plan having an AQL of 0.65 percent. Appropriate i and f values shall be as specified by the procuring activity. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.3.1, using equipment in accordance with 4.2.4.

4.2.3.3 Profile and alignment (see dwg. 8881026) - Major defect - Code No. 03009.-The cartridges, including those which have had the bourrelet repainted (see 3.5.4), shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.4 Strength of heat weld (see 3.7) Major defect, Code No. 04033.-One obturating band sample, minimum, after every 350 cartridges, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.3.4 using test equipment in accordance with paragraph 4.2.4. If the requirement as specified in 3.7 is not met, those 350 preceding cartridges represented by the tested sample shall be rebanded and retested.

4.2.3.5 Maximum outside diameter (see dwg. 8881026) Major defect, Code No. 04034.-The cartridges shall be gaged 100 percent for maximum outside diameter. Any cartridges which fail to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with paragraph 4.2.4.

4.2.3.6 Proving ground test.-The cartridges shall be observed for compliance with the requirements as classified in Table II.

TABLE II

	Defect	Defect Classification	Code No.
(a)	Failure to fire (see 3.4.1)	Major	04015
(b)	Safety (see 3.4.2)	Critical	04016
(c)	Impact functioning (see 3.4.3)	Major	04017
(d)	Horizontal range less than 80 percent of mean horizontal range but not less than 70 percent of mean horizontal range (see 3.4.4.1)	Special (see 6.8)	04021

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(e)	Horizontal range less than 70 percent of mean horizontal range.	Critical	04022
(f)	Range standard deviation greater than maximum lot values (see 3.4.4.2)	Major	04027

4.2.3.6.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, a sample of 80 cartridges shall be selected from each lot for this test. If a premature burst occurs (see 3.4.2) or if one or more cartridges fail to comply with the horizontal range requirement (see 3.4.4.1) or if two or more cartridges fail to comply with the requirement of 3.4.1 or if four or more cartridges fail to comply with the requirement of 3.4.3, the lot shall be rejected. The disposition of any lot, having cartridges with special defects, shall be made in accordance with paragraph 6.8. In addition, if the horizontal range standard deviation for the sample cartridges fired at charge zone one or charge zone nine multiplied by the factor 0.79, exceeds the applicable max. limit specified in Table I (see 3.4.4.2), the lot shall be rejected. This multiplication factor depends on the size of the sample. The test shall be performed as specified in 4.3.2.

4.2.3.6.2 After three consecutive lots have met the criteria of 4.2.3.6.1, a sample of fifty cartridges from each lot shall be tested. The lot shall be rejected if a critical defect occurs during any portion of the test or if two or more cartridges fail to comply with the requirement of 3.4.1 or if four or more cartridges fail to comply with 3.4.3. The disposition of any lot, having cartridges with special defects, shall be made in accordance with paragraph 6.8. In addition, if the horizontal range standard deviation, at charge zone one or charge zone nine multiplied by the factor 0.75, exceeds the applicable max. limit specified in Table I (see 3.4.4.2), the lot shall be rejected. This multiplication factor depends on the size of the sample. The test shall be performed as specified in 4.3.2.

4.2.3.7 Initial Production Test.-One-hundred-eighty sample cartridges from the first lot produced shall be selected for this test. The samples shall be grouped for test as specified in 4.3.3. Acceptance criteria shall be as specified in Table III.

TABLE III

Defect	Group (ADG) Acc	Group (ADG) Rej	Group (BEH) Acc	Group (BEH) Rej	Group (CFI) Acc	Group (CFI) Rej
Table II (a)	INFO		0	1	0	1
Table II (b)	0	1	0	1	0	1
Table II (c)	INFO		INFO		INFO	

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Table II (d)	Refer lot if one or more samples fails in this category.					
Table II (e)	0	1	0	1	0	1

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-8881026 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test Methods and Procedures

4.3.1 Determination of cavities.-The sample cartridges shall be subjected to x-ray examination. In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with applicable requirements. (Note.-Radiographic equipment, operations and procedures shall be qualified in accordance with Specification MIL-STD-453. In addition, the sensitivity of x-ray techniques shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.3.1.2). In the event an original x-ray picture leaves doubt as to acceptability, the cartridge shall be re-x-rayed at the angle 90 degrees from that of the original exposure and in the same plane.

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4.3.1.1 Check of sensitivity of x-ray technique.-The sensitivity of x-ray technique shall be determined, by the means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter including the hole in the center, is sharply defined on the x-ray film. Unless otherwise specified the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thicknesses and shall be equal to 2.0 percent of the max. equivalent thickness. The penetrameter shall have a hole the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Specification MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated by Figure 1 of this specification.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards be used (see 6.4). Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

4.3.2 Proving ground.-The test shall be performed at a Government proving ground in accordance with the applicable USATECOM Materiel Test Procedures. The cartridge shall be fired from an 81MM mortar for which the round is standard at an elevation of forty-five degrees. One half of the sample cartridges shall be fired with an ignition cartridge and one propellant increment (charge zone 1) and one half of the sample cartridges shall be fired with an ignition cartridge and nine propellant increments (charge zone 9). The individual velocities, max. pressure, fuze action and impact ranges shall be obtained and recorded. The standard deviation and the arithmetic average of the impact ranges shall be calculated and recorded. Hawk radar shall be used to observe for metal parts security and in the case of short rounds or unobserved dud rounds, radar magnetic tape records of such rounds along with the three preceding and three following rounds shall be forwarded for analysis to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND7.

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4.3.3 Initial Production test.-The samples shall be segregated into 9 groups of 20 rounds each identified as Group A through Group I and subjected to the following environments in the specified sequence.

4.3.3.1 Transportation-Vibration.-The cartridges in sample groups A through I in the standard Class A packing for the item (Ammo Containers and boxes) shall be subjected to a simulated vibration environment of 1000 miles of two wheeled trailer transportation plus 3 hours of aircraft transportation in accordance with TECP 700-700, Interim Pamphlet 70-73, at the following temperatures:

Groups A, B, C	-40°F
Groups D, E, F	+70°F
Groups G, H, I	+125°F

On completion of this test, the cartridges shall be x-rayed to insure that the fuzes are not armed.

4.3.3.2 Five Foot Drop Test.-After conditioning as specified in 4.3.3.1, the cartridges in sample groups A through I, packed in ammunition containers only, shall be subjected to a Five Foot Drop on a steel plate. Each round will be dropped once. The rounds shall be dropped in random orientation on the nose, 45° nose down, 45° base down, horizontally and base down.

4.3.3.3 Temperature and Humidity.-After conditioning as specified in 4.3.3.2 the cartridges in sample groups A through I, packed in the standard Class A packing for the item (Ammo Containers and boxes) shall be subjected to two 14 Day Jan Cycles of the Temperature and Humidity Test, Test 105 of MIL-STD-331. The samples shall then be conditioned as specified in 4.3.3.4 and fired as specified in 4.3.3.5.

4.3.3.4 Temperature conditioning.-Upon completion of the conditioning sequence specified, the cartridges in sample Groups A through I shall be temperature conditioned for a minimum of 24 hours as follows:

Groups A, D, G	-40°F (60 samples)
Groups B, E, H	+70°F (60 samples)
Groups C, F, I	+125°F (60 samples)

4.3.3.5 Ballistic test.-The cartridges shall be fired from an M29 Mortar at 45 degrees elevation at the conditioning temperature specified in 4.3.3.4 plus or minus 5 degrees F.

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Half the rounds in each group shall be fired at charge 0 and half at charge 9. Smear camera coverage shall be provided within 30 feet of the muzzle. Observation shall be made for muzzle velocity, maximum chamber pressure, range and dispersion, impact functioning and security of parts. The cartridges immediately prior to firing shall be visually and manually inspected for unusual conditions such as moisture within the container, looseness of primer, fin or fuze, difficulty in readying packed ammunition. All deficiencies should be noted. All rounds in the samples shall be fired except in the case of a round exhibiting a dangerous condition.

4.3.4 Strength of heat weld.-The sample shall be tested for integrity by applying the required load at the heat weld, radially and in a direction away from the axis (center) of the ring.

5. Preparation for delivery

5.1 Preservation and packaging

5.1.1 Level A (Fiber container).-The cartridges shall be preserved and packaged in accordance with dwg. 9230175.

5.2 Packing

5.2.1 Level A.-The cartridges in fiber containers shall be packed in accordance with dwg. 9230176.

5.3 Marking

5.3.1 Marking of the fiber container shall be in accordance with dwg. 9230175. The wood box shall be marked in accordance with dwg. 9230176.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs

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shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny arsenal design review will be accomplished normally within one month after receipt.

6.4 Film density.-"American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N.Y., has been found satisfactory as a source of information on film densities.

6.5 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridges, and shall not constitute a basis for rejection against

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either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.6. Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND

6.7 Clean scrap.-Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.8 Special defects

6.8.1 Definition.-A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

6.8.2 Disposition.-Upon investigation and analysis of lots exhibiting special defects by the responsible technical activity a recommendation of acceptance or rejection shall be made to Director of Quality Assurance, U.S. Army Munitions Command. Final acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U.S. Army Munitions Command.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1315-A-598

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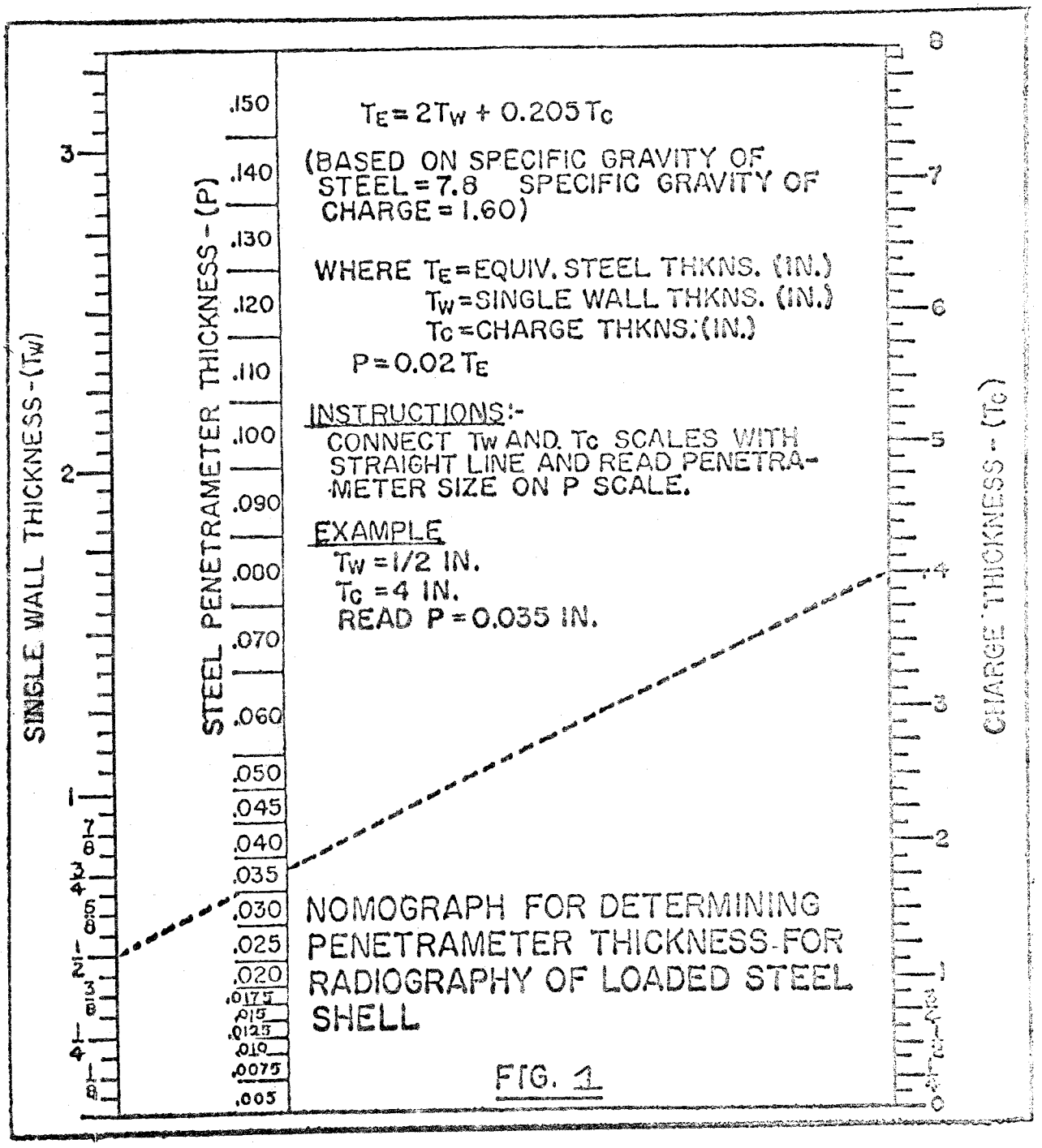


FIG. 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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